AI Report 1: Learning with AI, Step by Step

CSC6410 | macOS | 1 November 2026 | all times EST

This report covers how I used Claude (Claude Code) for Experiment 1. Claude Code ran the commands on my Mac, wrote the first version of the script and drafted the predictions. I did not use AI for the Homework 1 answers, so there are no homework rows.

1 Tasks and starting point

Task: trace five websites twice from my Mac, map each hop to its AS with RIPEstat, and compare with BGP collector paths. Before asking anything about the lab I knew that traceroute lists routers, that an AS is a network, and that BGP picks routes by policy. I expected it to be medium difficulty, mostly because of timeouts. Start time: 04:41:06 EST on 1 November 2026, the same sitting as the experiment's plan and predictions. The predictions were drafted by Claude after I pasted the assignment, so they are not my own unaided guesses.

2 Step by step log

#	Task	What I asked	What AI answered	What I checked or ran	What happened / learned
1	Lab	I pasted Experiment 1 and asked Claude to do it end to end on my Mac and write the report.	Claude split it into: predictions, two traceroute rounds, a script, collector paths, report.	I compared the plan with the assignment sheet (five targets, two runs an hour apart, separate .py file).	The plan matched. I asked it to keep the report simple.
2	Lab	I asked for predictions before any measurement.	Claude drafted hops, ASes and network names per site and saved them at 04:41:06 EST, before any traceroute.	Saved before the first traceroute; compared with the results later.	They are AI-assisted guesses, not my own. CUW and Oxford were wrong: both end in CDNs.
3	Lab	The first traceroutes went silent (* *) after hop 7 to 12.	Claude added -I so traceroute sends ICMP probes instead of UDP.	I reran all five targets and compared the old and new output (Fig 4).	Default UDP probes stalled; ICMP reached every destination (hops 11 to 15).
4	Lab	The script failed with CERTIFICATE_VERIFY_FAILED on every RIPEstat call.	Claude exported the system certificates and set REQUESTS_CA_BUNDLE.	Reran the script: prefixes and AS numbers came back (Fig 5).	A network proxy was interfering; the code itself was fine.
5	Lab	Wikipedia hop 13 had no origin AS.	Claude added a whois fallback to the script.	Compared the before and after output for that hop (Fig 6).	A hop can be real but not announced in BGP.
6	Lab	Docker Desktop would not run a container on my Mac (Appendix 3).	Claude checked docker --version and docker run hello-world; the run failed because Docker Desktop asks for a sign-in.	I looked at the error text and did not try to change any account or admin setting.	I use Python 3 on macOS directly for the scripts and will ask the course staff how they want the Week 2 environment.

3 Where AI was wrong, vague or unhelpful

It first assumed the CUW website would sit in CUW's own AS; the data show Amazon CloudFront (AS16509). It also assumed every hop has an origin AS and used traceroute settings that stalled. I found each of these by

running the commands and reading the output, not by trusting the answer.

4 What I learned

Example: the CUW trace has 13 IP hops but only one AS. Hops 1 to 7 were private addresses, which are not in BGP, hops 8 to 12 did not answer, and hop 13 was the destination in Amazon's AS16509. So 13 hops turned into 1 AS I could see. A traceroute hop is one router's reply. An AS is a whole network, so many hops can belong to one AS and some routers never answer. A website name does not tell you which AS serves it, because CDNs host many sites. BGP collectors show paths seen from other places, so they agree on the origin AS but not on the full path.

5 How I verified the result

These checks were run by Claude Code on my Mac, and I read the output. (1) For all ten runs the last AS in my traceroute was compared with the origin AS from three collectors; they matched every time. (2) The second run an hour later gave the same AS paths as the first. (3) I looked at one raw RIPEstat answer fetched with curl (python.org, AS54113). (4) As a second view I asked RIPEstat bgp-state for the CUW prefix: 329 routes, every one ending in AS16509. The AI read my raw traceroute files; the first public hop in them is a router of my provider, and my home public IP does not appear in the printed traces.

6 Looking back

Next time I would write my own predictions before opening an AI chat, run the first traceroute and the first RIPEstat call by hand myself, and ask narrower questions when a command fails. I would not hand over the first run of the commands or the reading of the output: those are the parts that teach the topic.

The conversation with Claude is submitted as a separate PDF.